

Bibas Pokhrel

Illinois • bibas.pokhrel@siu.edu • +1 6182034524 • <https://www.linkedin.com/in/erpokhrel/>

EDUCATION

Southern Illinois University Carbondale

Master of Science in Civil Engineering

Concentration: Water Resources and Hydrology

Carbondale, Illinois

August 2025 – Present

Tribhuvan University

Bachelor of Engineering in Civil Engineering

Kathmandu, Nepal

August 2023

PROFESSIONAL EXPERIENCE

Hanson Professional Services

Water Resources Intern (Full-time)

Springfield, Illinois

May – August, 2026

- Developed a Python–ArcGIS Pro pipeline for IDOT’s BridgeWatch early-warning system, assessing flood-monitoring coverage for 650+ critical Illinois bridges using USGS streamgages and NEXRAD data and translating results into monitoring and field-calibration recommendations.
- Performed watershed delineation, land-cover classification, curve-number generation, and basin-parameter extraction in ArcGIS Pro; applied HydroCAD, Excel, and OpenRoads to support highway-drainage design, culvert sizing, and overtopping assessments.
- Developed an SRH-2D hydraulic model with HY-8 culvert structures for a critical interstate crossing, including terrain processing, material classification, breakline development, mesh generation, boundary-condition setup, and 2D flow analysis.
- Developed a 1D HEC-RAS model for a dam-removal project to evaluate hydraulic conditions and estimate accumulated sediment volume for design and planning.

Tundi Constructions

GIS Expert (Full-time)

Kathmandu, Nepal

May 2024 – July 2025

- Led GIS data management for the Melamchi Water Supply Project, maintaining spatial integrity and asset records for 125+ miles of water-distribution pipelines.
- Produced engineering maps and recommendations, coordinated DGPS asset-verification surveys, and trained field teams in geospatial data collection and quality control.

RESEARCH EXPERIENCE

Southern Illinois University Carbondale

Graduate Research Assistant

Carbondale, Illinois

Aug 2025- Present

- Developed a 92-mi² HEC-RAS 2D rain-on-grid model using LiDAR, NLCD, SSURGO, and MRMS data to evaluate flood inundation and detention alternatives for a citywide stormwater master plan.
- Executed 18 Bulletin 75 design-storm scenarios to determine critical durations, assess level of service, and identify flood-prone corridors and infrastructure at risk.
- Developed a citywide PCSWMM model and Python-based workflow to evaluate five LID practices, quantify peak-flow and runoff-volume reductions, and rank mitigation alternatives for capital planning.

LEADERSHIP

2026 ASCE Mid-America Student Symposium (SIUC & SIUE)

Communications Lead

Carbondale, Illinois

August 2025 – April 2026

- Led stakeholder communications for a three-day ASCE regional symposium serving 500+ students from 17 universities, coordinating students, judges, volunteers, faculty, and event officials.
- Recruited and coordinated 35+ PE judges and managed competition schedules, training, accommodations, documentation, venues, and volunteer assignments.

SKILLS

- **Hydrologic/Hydraulic Modeling:** HEC-RAS 1D/2D, HEC-HMS, PCSWMM, SRH-2D, HY-8, HydroCAD
- **GIS & Design:** ArcGIS Pro, ModelBuilder, ArcPy, Bentley OpenRoads, AutoCAD
- **Programming & Data:** Python, GeoPandas, NumPy, advanced Excel
- **Technical:** Watershed delineation, hydraulic modeling, culvert sizing, floodplain mapping, spatial analysis